

EDUCATION

PhD	Aug. 2022 - May. 2027
<i>University of Colorado Anschutz Medical Campus, Computational Cell Biology</i>	<i>Aurora, CO</i>
BS	Aug. 2017 - May. 2021
<i>Maryville University of Saint Louis, Biochemistry and Data Science</i>	<i>Saint Louis, MO</i>

RESEARCH EXPERIENCE

PhD Research - Computational Cell Biology	Aug. 2022 - Present
<i>University of Colorado, Gregory P. Way, PhD</i>	<i>Aurora, CO</i>
Post-baccalaureate Research Assistant - Genomics	2021 - 2022
<i>University of North Carolina at Chapel Hill, Shawn Ahmed</i>	<i>Chapel Hill, NC</i>
Research Assistant - Genome editing	2020 - 2021
<i>Centro de Tecnologia Canavieira</i>	<i>Saint Louis, MO</i>
Undergraduate Research Assistant - Quantitative spectroscopy	2019 - 2020
<i>Maryville University of Saint Louis, Thomas Spudich, PhD</i>	<i>Saint Louis, MO</i>
Undergraduate Research Assistant - Cell Biology	2018 - 2021
<i>Maryville University of Saint Louis, Stacy L. Donovan, PhD</i>	<i>Saint Louis, MO</i>
Research Internship - Biochemistry	2018 - 2019
<i>Elemental Enzymes</i>	<i>Saint Louis, MO</i>

SCIENTIFIC APPOINTMENTS

Wellbeing and engagement committee, Department of Biomedical Informatics	2024 - Present
<i>University of Colorado Anschutz Medical Campus</i>	<i>Aurora, CO</i>
PhD student	Aug. 2022 - Present
<i>University of Colorado Anschutz Medical Campus</i>	<i>Aurora, CO</i>
Journal Club Committee, Cell Biology, Stem Cells, and Development	2023 - 2025
<i>University of Colorado Anschutz Medical Campus</i>	<i>Aurora, CO</i>
Volunteer	2023 - Present
<i>Clear Directions Mentoring</i>	<i>Aurora, CO</i>
Cell Biology Graduate Teaching Assistant	2023 - Present
<i>University of Colorado Anschutz Medical Campus</i>	<i>Aurora, CO</i>
Research Assistant	May. 2021 - May. 2022
<i>University of North Carolina at Chapel Hill</i>	<i>Chapel Hill, NC</i>
Teaching Aid – Chemistry/Cell Biology	2019 - 2021
<i>Maryville University of Saint Louis</i>	<i>Saint Louis, MO</i>
Lab technician training manager	2019 - 2021
<i>Maryville University of Saint Louis</i>	<i>Saint Louis, MO</i>
Biology and Maths Tutor	2017 - 2019
<i>Maryville University of Saint Louis</i>	<i>Saint Louis, MO</i>

PRESENTATIONS AND INVITED LECTURES

Invited speaker	Jan. 28, 2026
<i>Michael Johnson Seminar Series, Maryville University</i>	<i>Saint Louis, MO</i>
Oral presentation	Dec. 4, 2025
<i>Neurofibromatosis Young Investigator's Forum</i>	<i>Baltimore, MD</i>
Oral presentation	Oct. 27, 2025
<i>Society of Biomolecular Imaging and Informatics</i>	<i>Boston, MA</i>
Poster presentation	Oct. 26, 2025
<i>Society of Biomolecular Imaging and Informatics</i>	<i>Boston, MA</i>
Oral presentation	Oct. 24, 2025
<i>Cell Biology, Stem Cells, and Development Retreat</i>	<i>Estes Park, CO</i>
Poster presentation	Feb. 10, 2025
<i>Systems applications for cancer biology</i>	<i>Aurora, CO</i>
Oral presentation	Dec. 3, 2024
<i>American Society for Cell Biology Cell Bio conference</i>	<i>San Diego, CA</i>
Poster presentation	Oct. 18, 2024
<i>Cell Biology, Stem Cells, and Development Retreat</i>	<i>Breckenridge, CO</i>
Poster presentation	Aug. 27, 2024
<i>Center for Health Artificial Intelligence Retreat</i>	<i>Denver, CO</i>
Oral presentation	Jul. 18, 2024
<i>Computational Systems for Integrative Genomics</i>	<i>New York, NY</i>
Poster presentation	Dec. 3, 2023
<i>American Society for Cell Biology Cell Bio conference</i>	<i>Boston, MA</i>
Oral presentation	Aug. 19, 2023
<i>Center for Health Artificial Intelligence Retreat</i>	<i>Aurora, CO</i>
Poster presentation	Oct. 13, 2023
<i>Cell Biology, Stem Cells, and Development Retreat</i>	<i>Breckenridge, CO</i>

HONORS AND AWARDS

Neurofibromatosis Young Investigator's Forum Travel Award	2025
<i>Neurofibromatosis Young Investigator's Forum</i>	<i>Baltimore, MD</i>
SBI2 Travel Award, Yokogawa Electric Corporation	2025
<i>Society for Biomolecular Imaging and Informatics</i>	<i>Boston, MA</i>
Best Oral Presentation	2021
<i>Maryville University Research Conference</i>	<i>Saint Louis, MO</i>
Excellence in Biological & Physical Sciences	2021
<i>Maryville University</i>	<i>Saint Louis, MO</i>
Best Poster Award	2019
<i>Maryville University Research Conference</i>	<i>Saint Louis, MO</i>
Outstanding Junior Chemistry Student	2019
<i>American Chemical Society</i>	<i>Saint Louis, MO</i>

TEACHING AND MENTORING

Mentor - High School Student	Jan 2026 - Present
<i>University of Colorado Anschutz Medical Campus</i>	<i>Aurora, CO</i>
Guest Lecturer - CPBS 7601 Reproducible Computational methods course	Oct. 17, 2025
<i>University of Colorado Anschutz Medical Campus</i>	<i>Aurora, CO</i>

– Guest Lecturer - Leveraging AI in Cell Biology <i>Maryville University of Saint Louis</i>	Apr. 10, 2025 <i>Saint Louis, MO</i>
– Guest Lecturer - CPBS 7601 Reproducible Computational methods course <i>University of Colorado Anschutz Medical Campus</i>	Nov. 15, 2024 <i>Aurora, CO</i>

JOURNAL REVIEWER

– Ad hoc Reviewer - Review Commons	2025 - Present
– Ad hoc Reviewer - Cell	2024 - Present
– Ad hoc Reviewer - Molecular Biology of the Cell	2024 - Present
– Ad hoc Reviewer - BiorXiv	2023 - Present

PROGRAMMING SKILLS

- **Languages:** Python, R, SQL, Bash, Nextflow
- **Technologies:** HPC Orchestration (SLURM), Nextflow, Cloud (AWS, GCP), terraform, Docker, Git
- **Frameworks:** Pytorch, Scikit-learn, Pandas, NumPy, SciPy, Optuna, Seaborn
- **Visualization:** Matplotlib, Seaborn, ggplot2, Plotly, dash
- **Image Software:** CellProfiler, napari, Fiji/ImageJ
- **Core Skills:** Machine Learning, Deep Learning, Statistical Analysis, Data Visualization, Data Wrangling, Data Mining, database Management

PUBLICATIONS

Google Scholar page (with citation metrics):

<https://scholar.google.com/citations?user=mTdpDrwAAAAJ&hl=en>

- 7 **Michael J. Lippincott**, Jenna Tomkinson, Ibrahim Bilem, Mahomi Suzuki, Akiko Nakde, Toshiaki Endou, Simon Mathien, Felix Lavoie-Perusse, Carla Basualto-Alarcón, Gregory P Way. (2025) High-content live-cell time-lapse imaging predicts cells about to die via apoptosis. In Review at Cell Reports Methods. bioRxiv. <https://doi.org/10.1101/2025.10.23.684203>
- 6 Dave Bunten, Jenna Tomkinson, Erik Serrano, **Michael J. Lippincott**, Kenneth I. Brewer, Vince Rubinetti, Faisal Alquaddoomi, Gregory P. Way. (2025) Scalable data harmonization for single-cell image-based profiling with CytoTable. Patterns. <https://doi.org/10.1016/j.patter.2026.101514>
- 5 Erik Serrano, John Peters, Jesko Wagner, Rebecca E. Graham, Zhenghao Chen, Brian Feng, Gisele Miranda, Alexandr A. Kalinin, Loan Vulliard, Jenna Tomkinson, Cameron Mattson, **Michael J. Lippincott**, Ziqi Kang, Divya Sitani, Dave Bunten, Srijit Seal, Neil O. Carragher, Anne E. Carpenter, Shantanu Singh, Paula A. Marin Zapata, Juan C. Caicedo, Gregory P. Way. (2025) Progress and new challenges in image-based profiling. Molecular Systems Biology. <https://doi.org/10.1038/s44320-026-00197-7>
- 4 Abigail Mumme-Monheit, Grace E. Gustafson, Colette A. Hopkins, Raisa Bailon-Zambrano, Juliana Sucharov, **Michael J. Lippincott**, Gregory P. Way, Kathryn L. Colborn, James T. Nichols. (2025) A quadratic paradigm describes the relationship between phenotype severity and variation. Nature Communications. <https://doi.org/10.1038/s41467-025-63316-2>
- 3 **Michael J. Lippincott**, Jenna Tomkinson, Dave Bunten, Milad Mohammadi, Johanna Kastl, Johannes Knop, Ralf Schwandner, Jiamin Huang, Grant Ongo, Nathaniel Robichaud, Milad Dagher, Masafumi Tsuboi, Carla Basualto-Alarcón, Gregory P. Way. (2025) A morphology and secretome map of pyroptosis. MBoC. <https://doi.org/10.1091/mbc.E25-03-0119>

- 2 Srivastava H, **Michael J. Lippincott**, Currie J, Canfield R, Lam MPY, Lau E. (2022) Protein prediction models support widespread post-transcriptional regulation of protein abundance by interacting partners. PLOS Computational Biology. <https://doi.org/10.1371/journal.pcbi.1010702>
- 1 Lister-Shimauchi EH, McCarthy B, **Michael J. Lippincott**, Ahmed S. (2022) Genetic and Epigenetic Inheritance at Telomeres. Epigenomes. <https://doi.org/10.3390/epigenomes6010009>